



Short Installation and Operation Instructions

REFU sol 08K ... 23K

English



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REFU Elektronik GmbH, Marktstraße 185, D-72793 Pfullingen

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If you have any questions or comments regarding this manual, please contact: info@refu.com.

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REFU Elektronik GmbH assumes no liability in the event of damage as a consequence of the failure to observe the warning instructions in these operating instructions or of improper usage.

The operating, maintenance and safety instructions are to be read carefully prior to installation and commissioning.

The installation, commissioning and safety-related check must be undertaken by a qualified electrician.

Error-free and safe operation of the device requires proper and professional transport, storage, assembly, and installation, as well as careful operation and maintenance.

Only accessories and spare parts approved by the manufacturer are permitted.

Technical modifications to the device are not permitted.

Adherence to the safety regulations and provisions of the country in which the inverter will be used are to be observed.

For European countries, the valid EU guidelines are to be observed when using the inverter.

The environmental conditions, technical calculation data and connection conditions of the grid operator given in the product documentation must be upheld.

The technical data, calculation, connection and installation conditions are to be derived from the product documentation and must be observed under all circumstances.

No liability is assumed for damages in connection with force majeure and disaster situations.

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1 About this Short Installation and Operation Instructions

These operating instructions form part of the product.

- ⇒ Read the Short Installation and Operation Instructions carefully before installing and using the product.
- ⇒ Keep the operating instructions readily available with the device for the entire service life of the product.
- ⇒ Provide all future users of the device access to the operating instructions.
- ⇒ For more information on the device, troubleshooting, and options under:
www.refu.com

1.1 Symbols and Markup

	Prerequisite
⇒	One-step instruction
1.	Multiple-step instruction
•	Bulleted list
Highlighting	Highlighting within a text
	Result

1.2 Warning Notices

1.2.1 Layout of a Warning Notice

SIGNALWORD / WARNING INSTRUCTIONS CATEGORY

Type and source of danger

Consequences if the warning is not observed

⇒ Required action to avoid the risk

1.2.2 Categories of Warning Notices

NOTICE

Can lead to property damage if ignored.

Follow the warnings to avoid damage or destruction of the product!



CAUTION

May cause minor personal injury if ignored.

Follow the warnings to avoid injury!



WARNING

May result in death or serious personal injury if ignored.

Follow the warnings to avoid serious injuries!



DANGER

Leads directly to death or serious personal injury if ignored.

Follow the warnings to avoid death or serious injury!



This symbol represent important notes concerning potential harm to people, this unit, or associated equipment.

1.3 Information



Note:

A **notice** describes information which is important for the optimum and cost-effective operation of the equipment.

2 Safety Information

2.1 Intended Use

The REFUsol 08K ... 23K inverter, also referred to in these operating instructions as the inverter, is a solar inverter, which transforms the direct current generated by the PV generator (photovoltaic modules) into alternating current and feeds this to the public power supply network.

The inverter is manufactured according to the rules and current state of technology and takes into account the valid EU, IEC and UL guidelines respectively.

The inverter fulfils the requirements according to protection class IP 65 / NEMA 4 and can be used indoors and outdoors.

Any other use is not considered to be a designated use. The manufacturer accepts no liability for damages resulting from this.

2.2 Qualification of Personnel

Target group for these instructions are professionals who can assess the assigned tasks due to their professional training, knowledge and experience as well as knowledge of the relevant provisions and recognize possible dangers.

Electrical work must only be carried out by qualified electricians.

Only suitably trained and qualified personnel are allowed to work on this inverter. Personnel are regarded as being qualified if they are sufficiently familiar with the assembly, installation, and operation of the product as well as with all warnings and safety measures set out in these operating instructions.

2.3 Dangers Arising from Misuse



DANGER

Danger to life from electric shock

- ⇒ Device may only be installed and serviced by qualified specialist technical personnel.
- ⇒ Only class A modules according to IEC 61730 must be used.
- ⇒ The PV generator must be free of earth potential.
- ⇒ The PV generator must be designed in accordance with the technical data of the inverter.
- ⇒ Before connecting or disconnecting the DC plug always open DC switch.
- ⇒ Every power supply line must be equipped with an appropriate AC-disconnecting device.
- ⇒ A consumer must never be switched between an inverter and line protection switch.
- ⇒ Access to the shutdown mechanism must always be free.
- ⇒ Installation and commissioning must be carried out properly.



DANGER

Danger to life from electric shock

After the device has been switched off, the interior may still contain life-threatening voltage

- ⇒ Do not open inverter.
- ⇒ Discharge time is at least 15 minutes.



DANGER

Danger to life from high discharge current

- ⇒ It is essential to ensure an earthing connection has been established prior to connection to the supply current circuit.

2.4 Protection against Contact with Electrical Parts



DANGER

Danger to life, danger of injury due to high electrical voltage

- ⇒ Installation of the inverter must only be carried out by trained specialist personnel. In addition, the installer must be accredited by the responsible utility company.
 - ⇒ General assembly and safety stipulations relating to working on high current facilities must be followed.
 - ⇒ Before switching on, a check must be made to ensure that the plugs are firmly in place (locked).
 - ⇒ Before plugging or unplugging the DC plug, always disconnect the connection to the PV generator.
 - ⇒ The feeder must be isolated and secured against being switched on again before working at the AC grid.
-

2.5 Protection against Contact with Hot Parts



CAUTION

Danger of burns caused by hot surfaces on housings

At an ambient temperature of 45 °C, the upper part of the housing as well as the refrigeration unit can reach a surface temperature of more than 75 °C.

- ⇒ Do not touch the housing surface near to heat sources.
 - ⇒ Allow the device to cool down for 15 minutes before touching the surface of the device.
-

2.6 Protection during Handling and Assembly



⚠ WARNING

Danger of injury during improper handling caused by crushing, shearing, cutting, striking, and lifting

- ⇒ The weight of the inverter is 38.4 kg!
- ⇒ Follow the general setup and safety regulations for handling and assembly.
- ⇒ Use suitable assembly and transportation equipment.
- ⇒ Avoid trap and crush injuries by taking suitable precautions.
- ⇒ Only use suitable tools. Use special tools where this is prescribed.
- ⇒ Use lifting equipment and tools in a technically correct manner.
- ⇒ If necessary, use suitable protective equipment (for example, goggles, safety footwear, protective gloves).
- ⇒ Do not stand under hanging loads.
- ⇒ Remove any liquids onto the floor immediately to avoid the danger of slipping.

2.7 Please Note Before Starting Up

- In the event of installation in France, the device must be furnished with the warning sticker laid down by UTE C 15-712-1. The warning stickers are included in delivery.
- Problem-free and safe operation of the inverter is conditional upon due and specialised transportation, storage, assembly and installation as well as careful operation and maintenance.
- Only use accessories and spare parts approved by the manufacturer.
- Adherence must be ensured to the prescriptions and stipulations regarding safety of the country in which the inverter is to be used.
- The ambient conditions stated in the product documentation must be observed.
- Starting up is prohibited until the entire system meets the national regulations and safety rules regarding use.
- Operation is only permitted with adherence to the national EMC prescriptions for the present use case.
- The manufacturer of the equipment or machine is responsible for ensuring compliance with the thresholds required by the respective national regulations.
- For European countries, the EU Directive 2014/30/EU (EMC Directive) is to be observed when using the inverter.
- The technical data, connection and installation conditions are to be taken from the product documentation and must be observed under all circumstances.
- Switching off of the inverter must first be effected on the AC side via the circuit breaker. Then the inverter is to be switched off on the DC side, should maintenance work need to be carried out on the DC side.
- It is not necessary to switch off the inverter overnight, as the inverter switches off completely as soon as no DC voltage is present at the input. If no switch-off is effected, the inverter switches on automatically in the morning when the PV generator supplies sufficiently high voltage. This means that the maximum output is generated.

2.8 Additional symbols and warning notices on the inverter

	This symbol advises that the user handbook must be read before the device is used.
	Hot surface! The housing can get hot during use.
	Discharging time of the device longer than 15 minutes.
 WARNING Dual Supply	Do not work on this device before it is disconnected from both power sources (grid and PV generator).

2.9 Disposal



Dispose of the packaging and replaced parts according to the rules applicable in the country where the inverter is installed.

Do not dispose of the inverter with normal domestic waste.

The inverter conforms to RoHS. That means that the device can be taken to municipal disposal sites for household appliances.

REFU Elektronik GmbH takes the inverter back completely. Please contact the Service team!

3 Installation

3.1 Assembly Site Requirements



DANGER

Danger to life due to fire or explosion

- ⇒ Do not install inverter in areas with risk of explosion.
- ⇒ Do not install inverter on flammable materials. Uphold fire protection class F30 according to DIN 4102 resp. the NFPA regulation
- ⇒ Do not install inverter in areas where easily flammable materials are stored.
- ⇒ Under no circumstances are the cooling fins of the heat sink to be covered.



WARNING

Danger of accident

- ⇒ When installing and servicing, unrestricted and safe access to the devices must be ensured.

The following requirements of the installation site must be upheld:

- Only vertical installation is permitted as otherwise the cooling of the device cannot be guaranteed (convection cooling).
- Do not subject the inverter to any direct sunlight.
- A fixed wall or a metal construction is necessary for installation. The weight of an inverter is 38.4 kg. Relevant construction regulations must be observed.
- The device is to be mounted at an appropriate distance from combustible materials.
- Installation in non-ventilated ended cabinets and enclosures is not allowed because the unit in operation emits heat.
- To allow for the heat dissipation required, keep the following minimum distances from the ceiling and wall as well as from neighbouring devices:

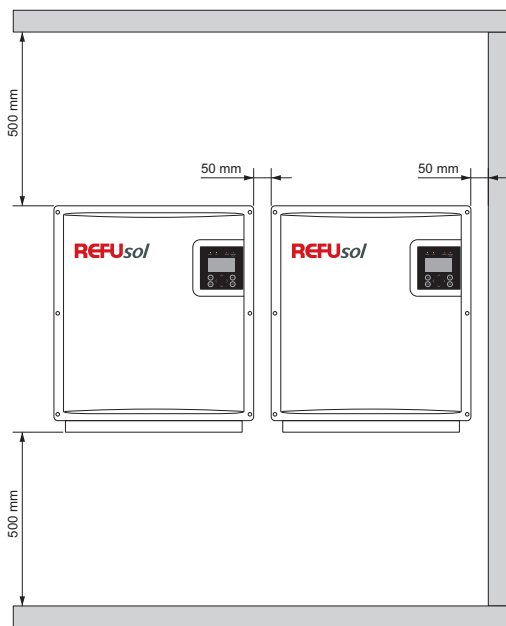


Fig. 1: Minimum distance

The device is best operated if the display is at eye level.

Protection class IP65 allows mounting in an outdoor area.

3.2 Transport

The devices must be transported under clean and dry conditions, if possible in their original packaging. The transport temperature must be between $-25 \dots +70 \text{ }^{\circ}\text{C}$. Permissible variations in temperature may not exceed $20 \text{ }^{\circ}\text{C}$ ($68 \text{ }^{\circ}\text{F}$) per hour.

3.3 Storage

The devices must be stored in clean and dry premises, if possible in their original packaging. The storage temperature must be in the range of $-25 \dots +55 \text{ }^{\circ}\text{C}$. Permissible variations in temperature may not exceed $20 \text{ }^{\circ}\text{C}$ ($68 \text{ }^{\circ}\text{F}$) per hour.

Note:



The inverter contains electrolytic capacitors which can be stored for no more than 1 year and at a storage temperature of $\leq 40 \text{ }^{\circ}\text{C}$ / $\leq 104 \text{ }^{\circ}\text{F}$ while in a de-energized state. If the storage time has been exceeded, please contact contact REFU Elektronik GmbH Service before connecting the inverter to your system!

3.4 Checking scope of delivery

The scope of delivery includes:

- Solar Inverter REFUsol 08K ... 23K
- Wall-mounting bracket
- Supplement with the following content:
 - 1x 5-pin contact insert, IP67, VC-TFS5-PEA for grid connection
 - 1x adapter housing IP67 VC-K-T3-R (10-25) PLOMB for grid connection
 - 2x cross-recessed flat-head screw, M5x20 => for mechanically securing the device in the wall bracket
 - 1x warning label in according UTE C 15-712-1 for installing in France

The screws for attachment to the wall are not included.

The IP67 VC-K-T3-R (10-25) PLOMB adapter housing allows sealing.

3.5 Unpacking the Device

NOTICE

Exposure to moisture, dust, or debris while unmounted in a non-vertical orientation poses a risk of ingress

⇒ Only unpack the device when it is to be installed. Failure to do so voids your warranty!

The inverters are heaviest at the top. They are therefore packed upside down to facilitate transport. You will therefore see the bottom side of the device (connectors) after having opened the package. Take the device at the two holding grips that are visible on the side and remove it from the packaging. When being unpacked, the device keeps the packaging grid locked in place on its housing. The packaging grid can be used to deposit the device on the floor. This prevents the cover from being damaged.



Fig. 2: Holding grip rear side positions

3.6 Installation

Check the delivery before installation (see page 15).



⚠ CAUTION

Danger of injury

- ⇒ Do not use the cover to hold the device. Only use the four holding grips to move the device.

NOTICE

Danger of damage to property

- ⇒ When designing the attachment of the wall-mounting plate, take the product weight of 38.4 kg into account.
- ⇒ Do not open device. Opening the device voids the warranty.

The inverter is mounted using the wall-mounting plate which is included in the scope of delivery.

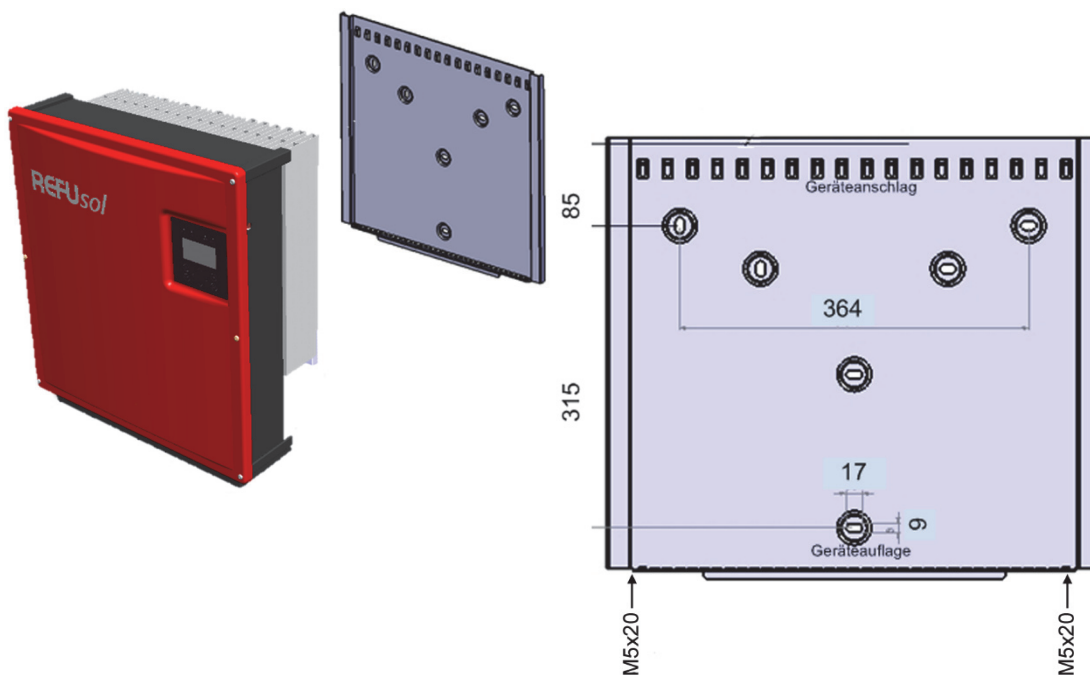


Fig. 3: Mounting the inverter

1. Use the wall bracket to mark the positions of the holes to be drilled.
2. Attach the mounting plate to the wall with the outer holes. The screws for attachment to the wall are not included. Screws with a diameter of 6 mm must be used.
3. Push the cooling fins of the inverter into the tabs of the wall mounting plate. Push the inverter upwards until it stops. Place the lower edge of the cooler onto the wall holder. Ensure that the rib profile is locked behind the nuts.

4. Secure the inverter in these nuts using the enclosed screws (M5x20). As an alternative, you can also use a padlock (shackle 4 mm in diameter) to protect the inverter against theft. The design of the wall bracket ensures that the inverter is automatically centered in this bracket.
5. In order to avoid adhesive residue on the display, remove the display protection immediately after installation.

3.7 Device connectors

The following figure shows the connectors of the inverter on its bottom side.

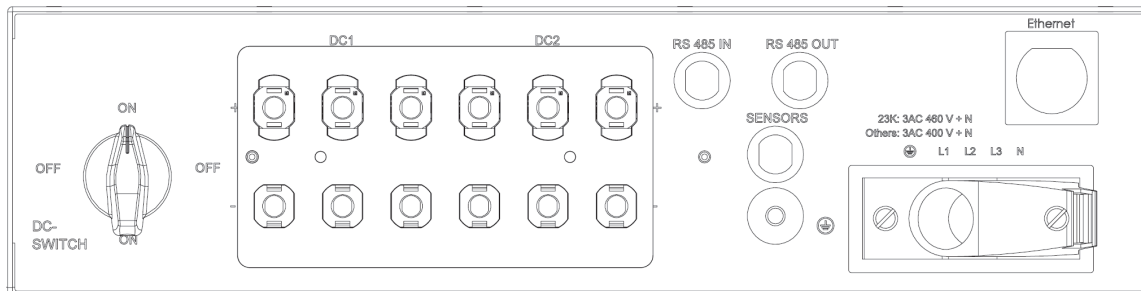


Fig. 4: Device connectors REFUsol 08K ... 23K

The inverter is provided with the following connectors, as seen from left to right, top to bottom:

- 6 pair of PV generator connections
- RS485 connectors (IN and OUT)
- SENSOR (connection: Irradiation and temperature sensor, external shutdown signal)
- Ethernet interface port
- Power connection

3.8 Grounding



⚠ DANGER

Danger to life from electric shock

- ⇒ The inverter must be earthed with the grounding bolt, otherwise a potential difference can build up, causing a risk of electric shock!

The inverter features a threaded bolt below the power supply port on the connection side for additional grounding. Grounding is intended to comply with the EMC requirements and to ensure optimum overvoltage protection. The ground wire cross-section must be bigger than the cross-section of the power supply line. In addition, ensure that the ground wire is placed as far away from and not directly in parallel to the power supply line.

The maximum permissible torque for this screw is 10 Nm / 7 ft lbs.

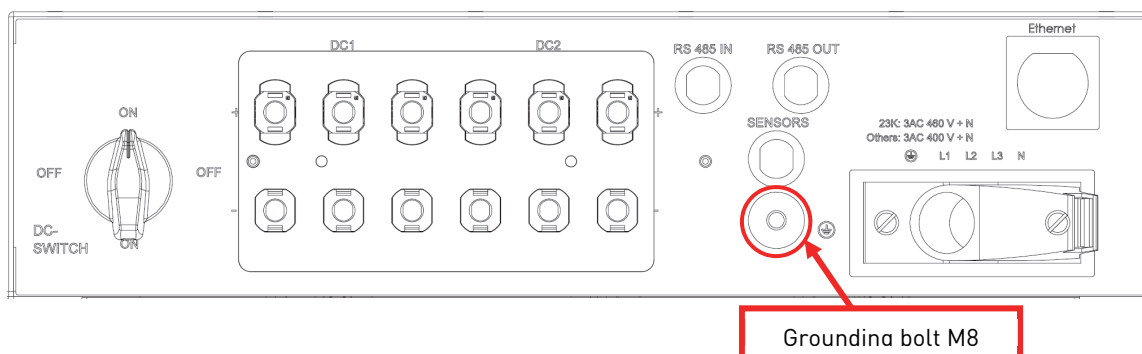


Fig 5: Grounding bolt

3.9 Residual current protection

Note:



The photovoltaic power supply inverters without transformers meet the fault protection requirements according to DIN VDE 0100-712, IEC 60364-7-712: 2002 and CEI 64-8/7 and can be operated with a type A residual-current circuit breaker without any functional impairment of the protection or the inverter.

The rated leakage current should be at least 300 mA per inverter.

3.10 Power Connection



⚠ DANGER

Risk of electric shock and fire caused by high discharge current

⇒ Before connecting the device to the supply circuit, establish a ground - connection by means of the labelled ground stud!

Note:



To ensure the IP65 protection class, only use the male and female connectors provided and connect them according to the connector manufacturer's mounting instructions. To prevent any penetration of moisture and dirt, unused inputs and outputs must be properly closed. Failure to do so could void your warranty!

The following mains systems are allowed:

- TN-C-Net
- TN-C-S-Net
- TN-S-Net

The connection to the power supply must be via a 5-wire line. For safety reasons, the PE protective conductor must always be connected.

The power supply line must be equipped with an appropriate line protection. More information regarding the power connection you will find in the Technical Data, page 43.

Always observe the following standards:

IEC 60364-4-41	Protective measures: protection against electric shock
IEC 60364-4-43	Protection for safety: protection against overcurrent
IEC 60364-5-52	Low-voltage electrical installations: selection and erection of electrical equipment

The respectively national standards are to be upheld.

The following specifications of the local grid operator are to be observed:

- Pertinent technical and special regulations
- Installation approval must be in place
- Check the line voltage. It must be within the technical data for that device.
- The AC side must not be measured by inserting test probes into the AC plug, because this will damage the plug contacts.

Note:



When using wire end ferrules with isolating collar, make sure you do not introduce the insulation of the wire end ferrule into the clamping area of the terminal.

3.11 Power Supply Line

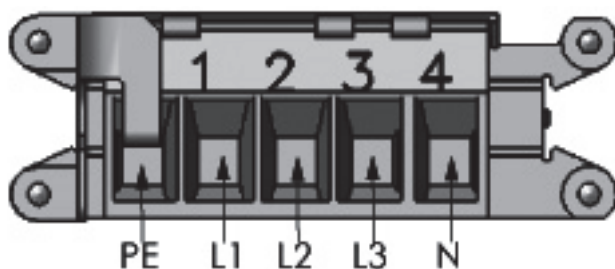


Fig. 6: Power Connection

Select the cross-section of the power supply line such that line losses are as low as possible. However, observe the following points:

- the installer must select the cable material according to the operation mode (installation location and laying type) and the national regulations.
- Due to the construction, the recommended feed line for all cross-sections is a fine wire.
- Copper wires must be used.
- Rigid wires are not recommended for the use of 16 mm² cables.
- The wires are to be installed in the correct position in order to minimise lateral forces on the net

connection. The lateral forces must not exceed 250 N.

- The seal is run through a hole in the screw (below the screw head of the adapter housing) and the opening provided and attached to the housing of the device.

The following table shows the maximum wire lengths upon use of a REFUsol 08K ... 23K depending on the cable cross section with a voltage drop below 1 %:

Line cross section	6.0 mm ²	10.0 mm ²	16.0 mm ²
Max. line length	30 m	50 m	70 m



Note:

In order to ensure IP65 protection, the AC connection housing supplied must be used.

3.11.1 Fitting of supplied power plug

The cable fitting of the standard connector housing supplied allows 5 x 6 mm² to 5 x 16 mm² cables to be connected. The maximum outside diameter of the power supply line may be 30.3 mm (e.g. Lapptherm 145, 5 x 6 mm²).

1. Bend cable into position.
2. Strip the cable as shown in the illustration.

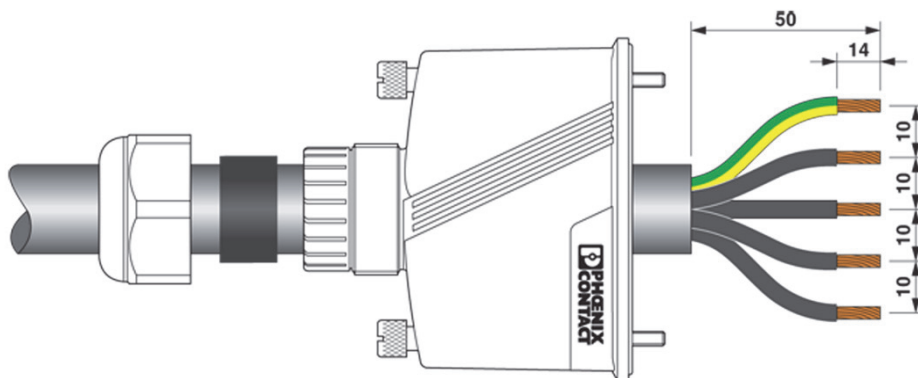


Fig. 7: Preparation of plug cable

3. Screw in plug.
4. Note torque specification according to illustration.

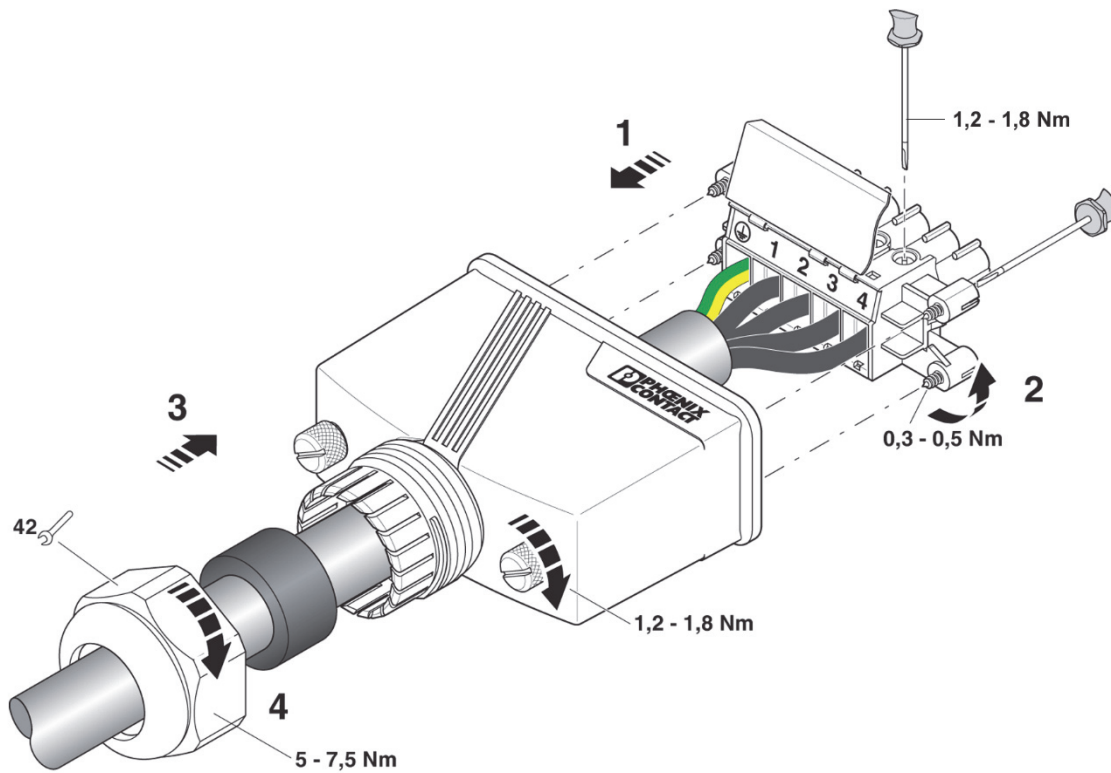


Fig. 8: Screw in plug

5. Determine diameter.
6. Seal connection.

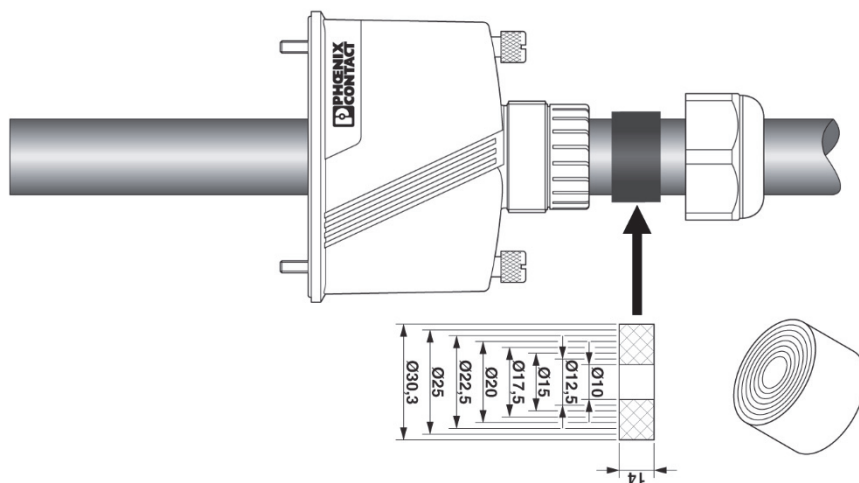


Fig. 9: Diameter

3.12 Grid Line Impedance

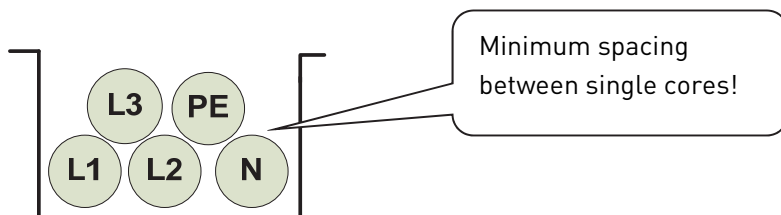
For better efficiency, large line cross-sections in single cables are increasingly used for power supply lines, especially if local conditions require long supply lines.

In large-scale plants, it is recommended as a best practice to locate the inverters in close proximity to the step-up transformer. If the inverters are located in a decentralized fashion, with long distances to the transformer, the higher AC cable inductance can lead to increasing line impedances. This can lead to voltage distortions and error messages with regards to:

- Grid overvoltage (long cable)
- Increased operating noise of the inverters (short cable)

To avoid these disadvantageous conditions, twisted lines should be used for power supply if possible. If laying of twisted lines is not possible, the following requirements must be met for single cores:

- Lay single cores next to each other.
- It is not allowed to lay single cores in closed, magnetically conducting materials (e.g., sheet steel pipe).
- If laid in open cable ducts, single cores should be laid such that the spacings between them are as small as possible.



Note:



To avoid unnecessary loss of energy yield, the sum total of the resistive and inductive voltage drop on the power supply line at nominal load should not exceed 1 % of the line voltage.

3.13 DC connection PV generator



⚠ DANGER

Danger to life due to high voltages from active PV strings

Before connecting the PV strings, connect the inverter to the power supply network and to the ground bolt to ensure that the device is **safely connected to the protective conductor.**

The connection of the PV strings to the inverter or the CombinerBox can only take place **without voltage**

Live PV strings can be under lethal voltages.

The maximum allowable voltage of 25 A per DC block must be upheld. The maximum DC voltage via all inputs must under no circumstances be exceeded.

- The DC connection is effected with Sunclix-plugs and sockets. More information see see “DC-Connection”, p. 24.
- Before connecting the PV strings an isolation measurement must be taken. Whenever it is switched on, the inverter automatically checks the insulation of the PV generator. If the isolation is defective, the inverter switches off automatically. The inverter can only be started once the PV generator isolation error has been removed.
- Be absolutely sure to verify proper polarity when connecting the PV strings. Any inappropriate connection of individual strings may damage the PV generator.

Protect the connectors such that they cannot be pulled off inadvertently.

The PV generator may not exceed the following operational characteristics under any circumstances!

Device type	REFU sol 08K ... 23K
Max. DC voltage at each input (USC)	1000 V
Max. voltage per DC block (DC1/DC2)	25 A
Max. DC current via all DC inputs	see “Technical data”, p. 43



Note:

Do not ground the positive or negative pole of the PV generator. Otherwise, the modules may experience a loss of power

NOTICE

Reverse currents are fault currents that occur in PV systems comprising parallel strings. Given short circuits of individual modules or cells in a module, the open circuit voltage of the string in question can drop so far that the intact parallel strings will drive a reverse current through the defective string. This may result in significant heating and therefore to destruction of the string.

What is more, the reverse current may cause secondary damage.

As no separation of the strings takes place within the inverter, each string must be separately protected by a string fuse connected in series with the other string fuses. In the event of a failure, this isolates the string from the intact strings so that destruction is prevented.

3.13.1 DC-Connection

Please note the following information (plug type, cross section) regarding the DC power cable:

Designation	Type	Art. No. PHOENIX CONTACT	Art. No. REFU	Outer Diameter	Cable cross section
Connectors +	PV-CM-S 2.5-6 (-)	1774687	922006	5 – 8 mm	4 – 6 mm ²
Connectors -	PV-CF-S 2.5-6 (+)	1774674		5 – 8 mm	4 – 6 mm ²
PV-C PROTECTION CAP		1785430	922007		

If necessary, the following photovoltaic Y distributors may be used:

Type	Art. No. PHOENIX CONTACT
PV-YC 6/ 1-0,12-S03 (-/++)	1787726
PV-YC 6/ 1-0,12-S03 (+/--)	1787739

For the adjustment of MC4 to Sunclix use the following adapter cable:

Typ	Type no. REFU sol	Type no. PHOENIX CONTACT
PV-AS-MC4/6-150-MN-SET	922008	1704982

NOTICE

In order to ensure IP65 protection, plug connectors and power supply connection cables must be matched to each other and all unused connectors shall be fitted with blanking plugs.

Only the original components by PHOENIX CONTACT can be used!

Failure to do so could void your warranty!

3.13.2 Connect PHOENIX CONTACT connector (SUNCLIX)



DANGER

Danger to life due to high voltages from active PV strings

- ⇒ The SUNCLIX connectors are only to be connected by electrotechnically trained personnel.
- ⇒ Never connect or disconnect SUNCLIX connectors when under load.

**⚠ CAUTION****Danger of injury or damage to property**

- ⇒ Only use the SUNCLIX connectors together with a PV1-F type 2.5-6 mm² solar cable or UL-certified solar cable (ZKLA stranded copper wire, AWG 10-14). A safe electrical connection is only guaranteed with this cable. Further cable types can be requested from PHOENIX CONTACT.
- ⇒ When laying the solar cable, observe the bend radii specified by the manufacturer.
- ⇒ Only connect the connectors to other SUNCLIX connectors.
- ⇒ When connecting, always observe the specifications for rated voltage and rated current. The smallest mutual value is valid.
- ⇒ Protect the connector against moisture and dirt.
- ⇒ Do not submerge the connectors under water.
- ⇒ Do not lay the plug directly on the roofing.
- ⇒ Protect unplugged connectors with a protective cap (e.g. PV-C PROTECTION CAP, 1785430).

3.13.2.1 Connect cable

1. Strip the cable by 15 mm with a suitable tool. Ensure no individual strands are cut off.
2. Carefully add stripped conductors with drilled bunched conductors until impact. The bunched conductor ends must be visible in the spring.
3. Close spring. Ensure that the spring is locked in place.
4. Push insert into the barrel.
5. Tighten the cable screws with 2 Nm.
6. Bring together plug and bush. The connection latches.
7. By pulling the coupler, check the correct connection.

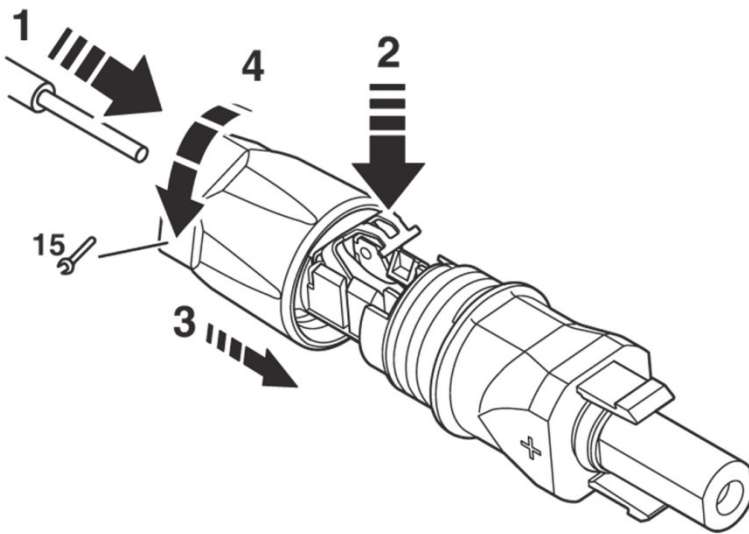


Fig. 10: Connect cable

3.13.2.2 Disconnect connector

A slit screwdriver is required with 3 mm wide blade (e.g. SZF 1-0,6X3,5, 1204517).

1. Enter screwdriver as demonstrated in picture below.
2. Leave screwdriver attached and separate bush and plug from one another.

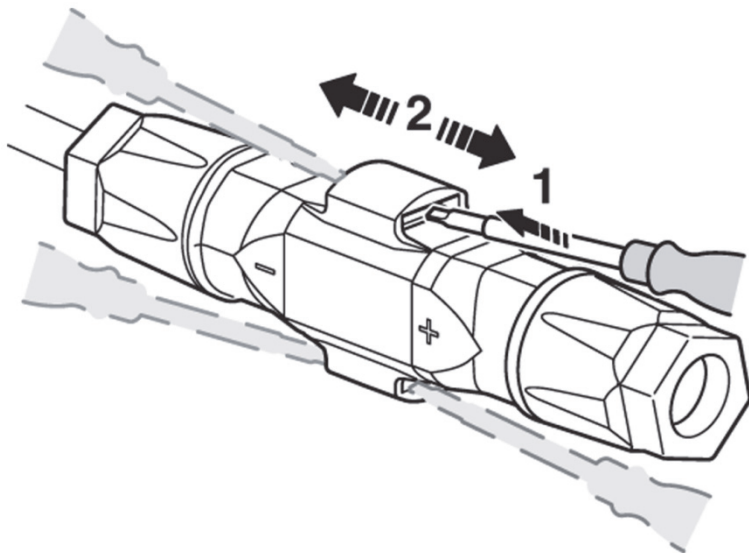


Fig. 11: Disconnect connector

3.13.2.3 Remove cable

1. Unscrew cable screwing.
2. Enter screwdriver as demonstrated in picture below.
3. Pry open connection and separate barrel and insert from one another.
4. Open spring with screwdriver. Remove cable.

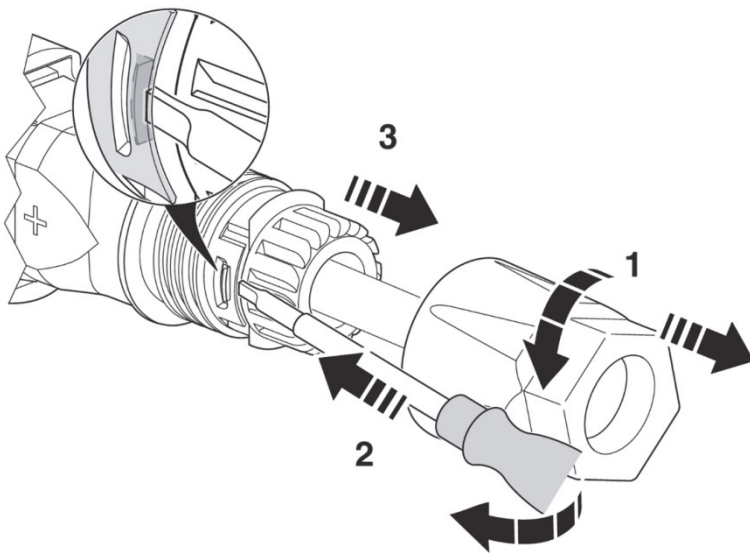


Fig. 12: Remove cable

3.14 Interface Port RS485

The RS485 interface supports the USS (Universal Serial Interface Protocol) and Modbus/Sunspec protocols, which can be used for transmission of data, for example, to a data logger of a remote monitoring system.

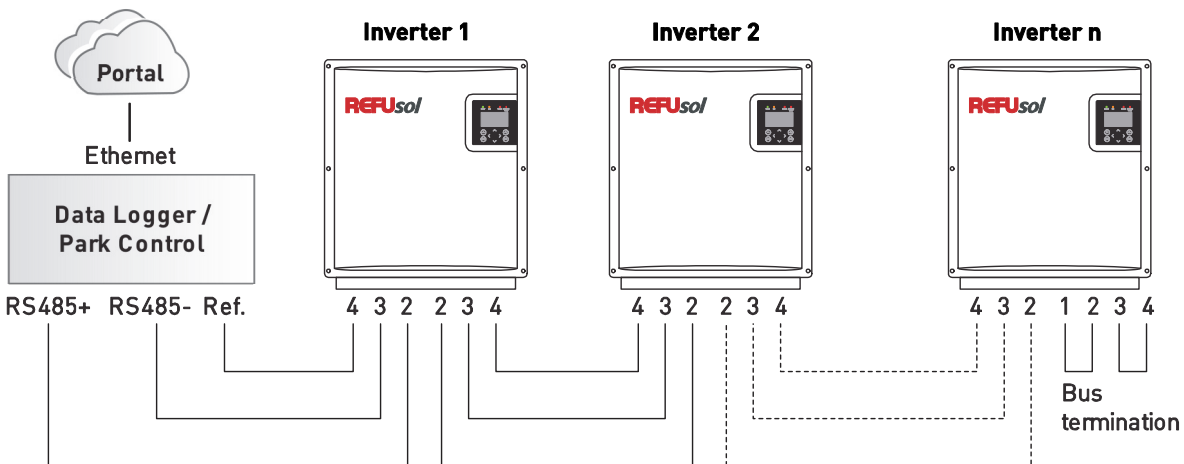


Fig. 13: Standard interface

When using this interface, please note that each device using the bus requires a unique address.

The bus termination is made by means of wire jumpers on RS485 OUT to the last bus user (inverter “n”). To make the termination take a short cable, remove the cable insulation, and connect the pin 1 to pin 2 as well as pin 3 to pin 4.

Pin	Signal
Pin 1	(only used for termination)
Pin 2	RS485 +

Pin 3	RS485 -
Pin 4	Reference

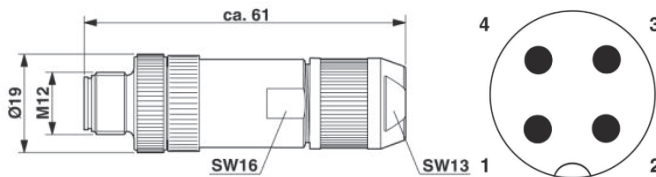


Fig. 14: Connector M12 x 1 straight, shielded; pole arrangement: male M12, 4 pins, A-coded, view of male connector side

NOTICE

In order to ensure IP 65 protection and the required and declared conformity with the domestic EMC standard a PHOENIX CONTACT type M12MS SACC-4SC SH plug and a shielded cable must be used. The outer diameter of the connecting cable can be max. 8 mm.

Failure to observe this can lead to damage to the inverter and therefore void the warranty.

The plug pair can be ordered at REFU Elektronik GmbH with Art. No. 922001.

3.15 Ethernet Interface Connection

The Ethernet interface supports Modbus TCP/IP protocol, and is Sunspec compliant, which can be used for transmission of data, for example, to a data logger of a remote monitoring system. Daisy-chaining of devices is not supported. Each inverter must be connected to an ethernet switch, in parallel.

Please use an Ethernet cable with S/FTP design (shielded foiled twisted pair) and Phoenix Contact plug type Quickon VS-08-RJ45-5-Q/IP67.

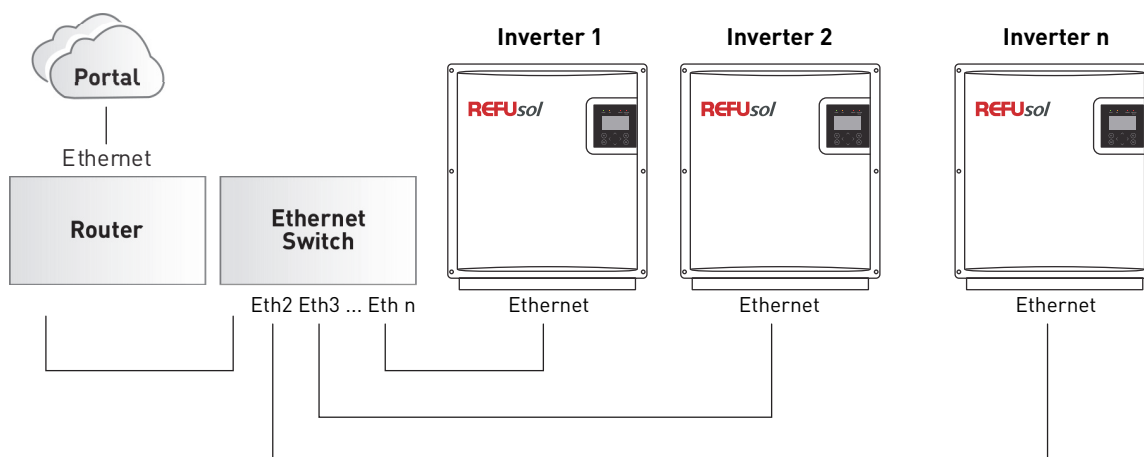


Fig. 15: Ethernet interface

NOTICE

In order to ensure IP 65 protection, the plug type mentioned above must be used. Failure to observe this can lead to damage to the inverter and therefore void the warranty.

The plug can be ordered from REFU Elektronik GmbH with Art. No. 922002.

4 Commissioning

Before commissioning the inverter, be sure the following steps have been completed:

- ☑ Confirm the correct power supply connection
- ☑ Confirm the correct connection of PV strings
- ☑ Confirm that connectors are protected such that they cannot be pulled off inadvertently



DANGER

Danger to life from electric shock

- ⇒ Before switching on the device, check whether the plugs are securely fitted (locked).

Do not pull off the connectors of the PV generator before you have met the following requirements:

- ⇒ Make the power supply line absent of voltage.
- ⇒ Make the device absent of voltage using DC circuit breaker.
- ⇒ Check that the DC cables of the PV generator are de-energized.
- ⇒ Protect the voltage supply from being reactivated



DANGER

Risk of electric shock and fire caused by high discharge current.

- ⇒ Before connecting the device to the supply circuit, establish a ground connection.

4.1 Turning on the Device

1. Verify that the device is connected to line voltage. If not, insert the external power fuse or turn on the circuit breaker.
2. Apply voltage by turning on the DC switch. The inverter only starts if there is sufficient voltage available from the PV panels.



Note:

The control panel, including its status indicators, display, and operator keys, is only active when the PV generator is supplying sufficiently high voltage.

4.2 Setting the Country Code and the Menu Language

When switching on for the first time an initial configuration is necessary.

The country code defines the country-specific grid monitoring parameters. The menu language is automatically set when the country code is selected. The menu language can subsequently be selected as

desired at any time, independent of the country abbreviation set in the menu. The country code is not set on delivery.

NOTICE

The selected country code can only be changed by service personnel.

After the initial setting and confirmation of the country code the country code can only be altered within 40 hours. This is also valid for replacement devices. The country code can then only be changed by authorized service personnel.

NOTICE

Cancellation of the operating license!

Operating the REFU**sol** inverter with the wrong country code can lead to the withdrawal of the interconnection agreement from the energy supplier. We assume no liability for the consequences of an incorrectly set country code!

4.2.1 Set Country Code/Grid Guideline

The following window appears on the display after initial switch on of the DC voltage and prompts you to set the country code. You can choose from among the offered countries.



1. Use the **^** and **v** keys to select the country code which is specific for your country and your location. The menu language is selected simultaneously with the country code. However, the menu language can later be changed independently of the country code.
2. Press **OK** to confirm. Then the grid feed guideline given by the energy supplier must be selected.



3. Select valid grid feed guideline with **^** and **v** buttons.
4. Press **OK** to confirm.

4.2.2 Accepting the Country Code

To be certain, you will be asked whether you wish to accept the country code. After accepting the country code it can only be changed within 40 hours.



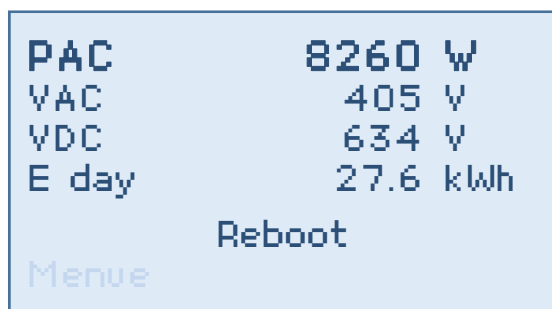
Only confirm the country code if you are absolutely sure it is correct.

If you are not sure, press **ESC** to cancel your selection and correct the selection.

If the country code is to be accepted, then confirm with **OK**.

A restart then takes place:

➤ "ERROR" and "GROUND FAULT" status LEDs flash red alternately.



4.3 Device Start

- ☒ Solar modules are irradiated with sufficient sunlight.
- ☒ Country code is set.
- ☒ Time and date is set.

The following procedure follows:

Self-test	All status LEDs are lit for approx. 6 seconds
Check	"START" status LED lights orange.
Activation	"START" status LED lights orange/LED "ON" flashes green.
Feed in	"ON" status LED lights green/normally all other status LEDs are off.

4.4 Control Panel

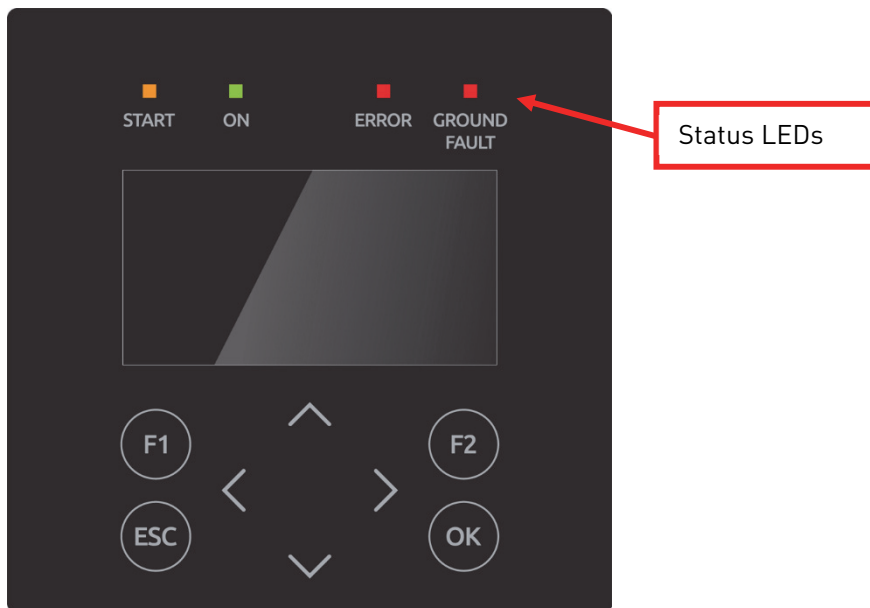


Fig. 16: Control panel with status LEDs, display and keys

F1	Displays the Menu
F2	In basic display: Reboot of display "Energy / x kWp" display: Input of standardized (normalized) data
◀▶	Function in the menu: Navigation within the menu level (previous menu, next menu). Function while parameters are edited: digit to the left, digit to the right (decade jump)
^v	Selection within the menu.
ESC	Acknowledge failures and exit from menu level, exit from input menu without entering data
OK	Confirm the selected menu (next menu level) and entered data.

4.5 Basic Screen

PAC	8260 W
VAC	405 V
VDC	634 V
E day	27.6 kWh
Feeding	
Menu	

Fig. 17: Operating mode display

PAC = Current feed power in Watts (W)

VAC = Line voltage in Volts (V)

VDC = Solar cell voltage in Volts (V)

E day = Energy yield of the day in kWh

4.6 Graphical Display

Press the **◀** arrow key once to display the development of the day's feed power.

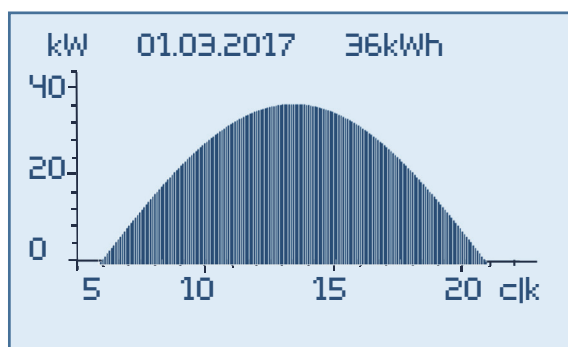


Fig. 18: "Today's" feed power display

Press the **▼** arrow key to display the development of the previous days.

Press the **ESC** key to return to the basic screen display.

4.7 Energy Yield Data Display

Press the **▶** arrow key to display the current yield data and the operating hours having currently elapsed.

Energy absolute	
Day :	0.7 kWh
Month :	5.3 kWh
Year :	49.2 kWh
Total :	265.8 kWh
Oper. hr :	16.1 h

Fig. 19: Energy yield absolute display

4.8 Standardized Energy Yield Data Display

Press the **▶** arrow key, then the **▼** arrow key to display the development of standardized energy yield data.

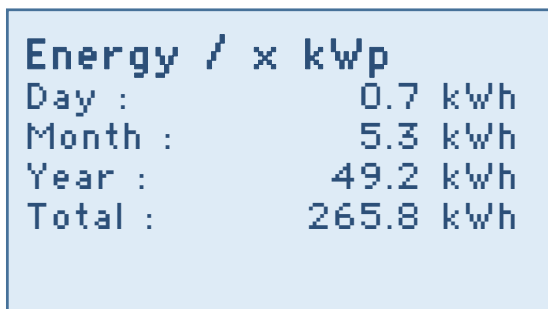


Fig. 20: Energy yield normalized on the display

Press the **ESC** key to return to the basic screen display.

4.9 Input of Standardized Data

To change the normalization, press the **F2** key and enter the currently connected PV generator power as follows:

- ◀▶ keys: Press the ◀ key to select the digits before the point.
 Press the ▶ key to select the digits after the point.
- ▲ key: Press the ▲ key to increment the selected digit by 1.
- ▼ key: Press the ▼ key to decrement the selected digit by 1.

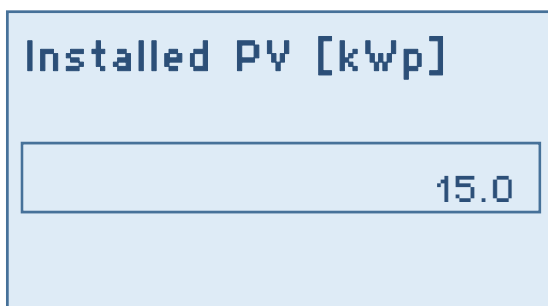


Fig. 21: Standardized data input display

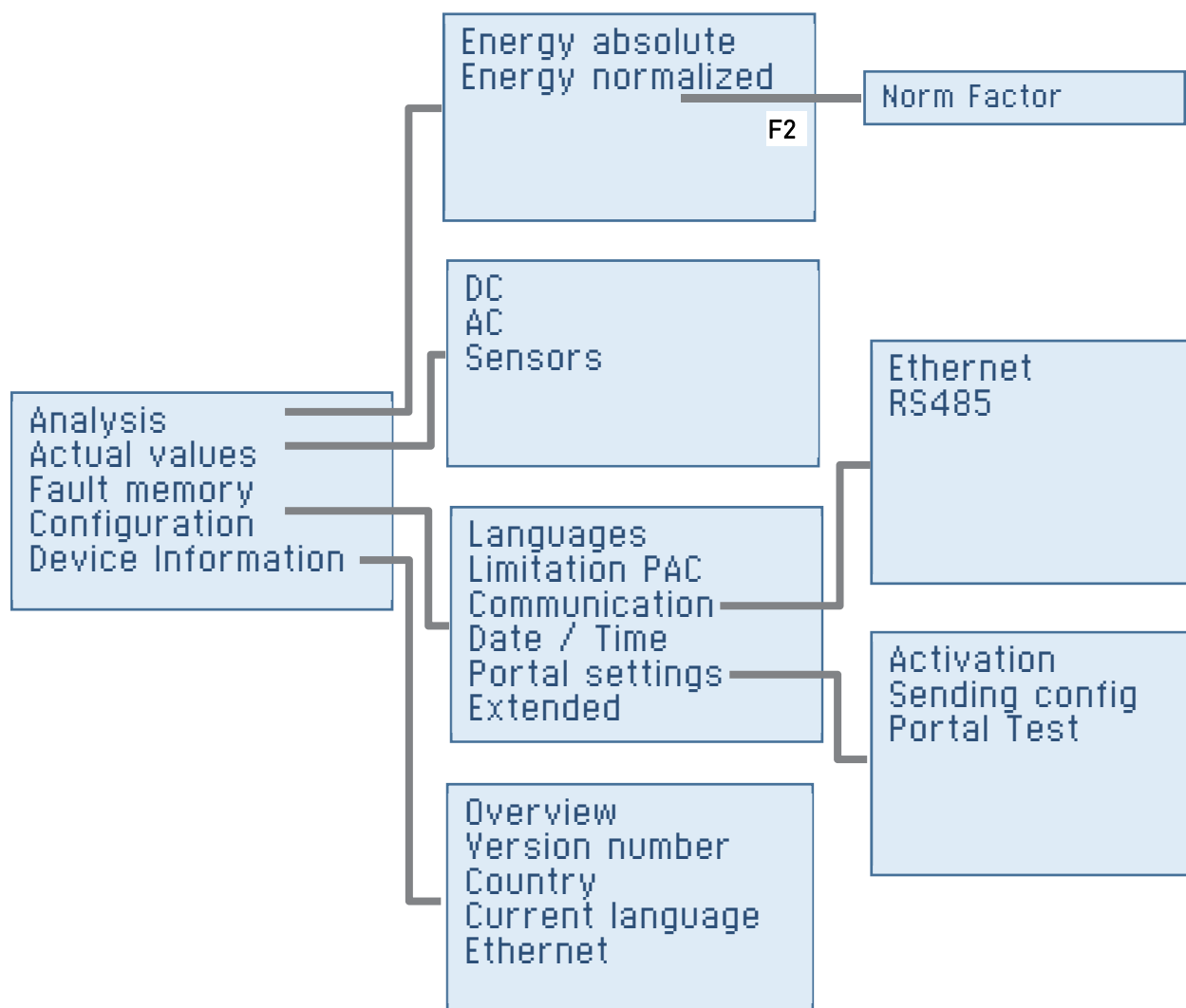
Press the **OK** key to apply the set value.

Press the **ESC** key to discard the value and display the previous “normalized yield” level.

Press the **F1** key to display the menu.

4.10 Menu Structure

The menu structure serves as a support to change to the individual information displays and setting displays.

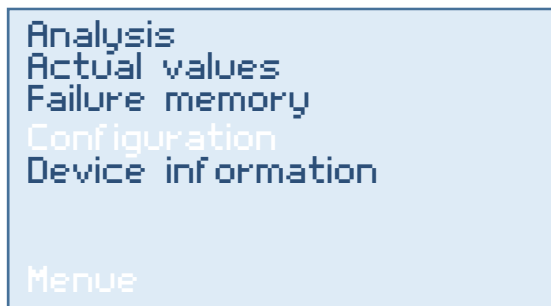


5 Configuration

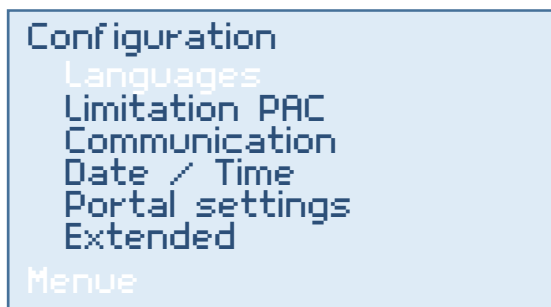
5.1 Changing the Menu Language

The language selection has no effect on the country code. Proceed as follows to change the menu language:

1. Press **F1** to open the menu.
2. Use the **▼** and **▲** keys to select the fourth menu item "Configuration".



3. Press **OK** to confirm.
4. Use the **▼** and **▲** keys to select the first menu item "Languages".



5. Press **OK** to confirm.
6. Use the **▼** and **▲** keys to select the desired menu language.
7. Press **OK** to confirm.
⇒ The menu switches to the selected language. The display is initially empty.
8. Press **ESC** to return to the menu.

5.2 Limitation of the Power Output

Proceed as follows in order to limit the power output of the inverter:

1. Using the **F1** key, select the menu item "Configuration" and confirm with the **OK** key.
2. Select the sub-menu "PAC Limit" and confirm with the **OK** key.
3. Enter the invert power output as percentage of the nominal power and confirm with the **OK** key.

⇒ The power limit is immediately applied.

5.3 Monitoring with REFUlog

The REFU**sol** 08K ... 23K inverter can be connected to the monitoring portal REFUlog using direct Ethernet connection to the internet, or via RS485 and the REFUcontrol data-gateway.

Once an internet connection is established, the inverter will automatically start to send data to REFUlog every 5 minutes. The sending interval can be changed with the configuration tool REFUset.

To view the inverter data, open www.refu-log.com with your browser and login or register as a new user.

You can use the activation code provided on the type label of the inverter to assign one or more inverter to a PV system.



5.4 REFUset – PC Configuration Tool

You can upgrade the firmware of the inverters and change country-specific parameters of the inverter with the PC Software REFUset. You can connect the PC with the inverters via Ethernet or USB using a RS485-USB converter.

For extended settings, you will receive a personal password on request from the REFU Service.

Functions

- Firmware update
- Export of inverter data
- Feed-in conditions voltage and frequency
- Ramp starting time in the event of a grid error
- Average voltage monitoring
- Phase conductor voltage monitoring
- Frequency-dependent power reduction
- Actual voltage monitoring (rapid disconnection)
- Actual frequency monitoring
- Reactive power ($\cos \varphi$)
- Power-up time

- K factor (fault ride through)

REFUset can be downloaded from www.refu.com.

5.5 Communication via Ethernet

The settings for the Ethernet communication can take place either manually or automatically (via DHCP). Manually is the default setting.

5.5.1 Automatic Setting via DHCP

1. Navigate to Menu **Configuration > Communication > Ethernet > DHCP**.

2. Enter "1" in DHCP input field.

3. Press **OK** to confirm.

4. Restart inverter by opening and re-closing the DC switch.

✎ The settings for IP address, subnet mask, standard gateway and DNS are automatically taken from the DHCP server, e.g. the router.

5.5.2 Manual Setting

In order to ensure a trouble-free operation, all values must be given by the responsible network administrator and individually set in each inverter.

1. Navigate to **Configuration > Communication > Ethernet > IP address**.

2. Enter IPv4 address. The IPv4 address is arbitrary, but must be consistent throughout the subnet.
Default value: **192.168.130.30**

3. Press **OK** to confirm.

4. Navigate to **Configuration > Communication > Ethernet > Subnet mask**.

5. Enter IPv4 mask. The IPv4 mask is arbitrary, but must be consistent throughout the subnet
Default value: **255.255.0.0**

6. Press **OK** to confirm.

7. Navigate to **Configuration > Communication > Ethernet > Standard gateway**.

8. Enter the address of the router that allows connection to the internet
Default value: **192.168.1.1**

9. Press **OK** to confirm.

10. Navigate to **Configuration > Communication > Ethernet > DNS**.

11. Enter arbitrary IPv4 address of the superior DNS server
Default value: **0.0.0.0**

12. Press **OK** to confirm.

13. Restart inverter.

5.6 Communication via RS485

For monitoring via RS485, you need to use REFUcontrol device to forward data to the monitoring portal REFUlog.

You need to assign an unique USS address to each connected inverter.

Addresses: 1 to 31

This address is required for communicating with the inverter via RS485.

⇒ After entering the address, restart the inverter to activate the new address.

Protocol type:	1: REFU USS protocol (preset) 2: (not used) 3: Meteocontrol USS 4: Modbus RTU Sunspec (address 0 cannot be used – address 0 is automatically set to address 1!) 5: Multimode USS / Modbus RTU Sunspec (in Multimode, you cannot use USS address 0 and 2!)
Baud rate:	57600 (preset and recommended) 115200 (for short RS485 networks)
Parity:	even
Handshake:	no
Data bits:	8
Stop bits:	1
Protocol:	REFU USS protocol

5.7 Portal Monitoring

To ensure the inverter sends data via Ethernet to the REFUlog monitoring portal, the portal monitoring must be active.

Open Menu Configuration > Portal Monitoring and enter the following value:

0 = Portal monitoring via Ethernet not active

1 = Portal monitoring via Ethernet active



Notice

If the inverter is connected with RS485 to REFUcontrol, the setting 0 (Portal Monitoring via Ethernet not active) must be used!

5.8 Sending Config

Menu Configuration > Portal monitoring > Config sending

The configuration data is required by REFUlog in order to recognise the inverter. If you want to send it manually, select the menu and set the parameter to 1. If the sending was successful, the parameter will be set back to 0 automatically.

5.9 Portal Test Function

Menu Configuration > Portal monitoring > Portal Test

On request of our service, you can send a test data package to the portal. Open the menu and select „yes“.

6 Maintenance

6.1 Maintain the Inverter

The cooling of the inverter is exclusively through natural convection, therefore the cooling system is maintenance free.

Periodically, based on site conditions, visual inspection of the cooling fins is recommended to identify any debris accumulation that could occur. If found, the debris must be removed to ensure that the convection cooling is not inhibited.

Cleaning with high-pressure cleaners is prohibited.

Annually, the DC switch has to be operated without electricity 5 times.

7 Technical data

7.1 Inverter

Type	08K	10K	13K	17K	20K	23K-MV
Item no.	867P008	867P010	867P013	867P017	867P020	867P023
DC data						
Recommended max. PV power (kW)	9.9	12	15.6	20.4	24	27.6
MPPT range (V)	370-850	410-850	480-850	460-850	490-850	575-850
DC start voltage (V)	350					
Max. DC voltage (V) (U _{sc_PV})	1000					
Max. DC current (A)	23	25	31.1	38.3	41.8	41
MPP tracker	1					
Number of DC connections	6					
DC disconnection switch	Yes					
Max. total short circuit current of the PV plant (I _{sc_PV}) (A)	50					
AC data						
Rated AC power (kW)	8.25	10	13	17	20	23
Max. AC apparent output (kVA)	8.25	10	13	17	20	23
AC grid connection	L1, L2, L3, N, PE					
Rated power factor/range	1 / 0.8i ... 0/8c					
Rated voltage AC (V)	400					460
Voltage range AC (V)	320-460					368-529
Rated frequency/frequency range (Hz)	50, 60 / 45...65					
Max. AC current (A)	3 x 12	3 x 16	3 x 21	3 x 29.2		
Max. distortion factor THD	2.5%			1.8%		
Max. efficiency	98.1%			98.2%		98.3%
Feed-in starting at (W)	50					
Internal consumption in night operation (W)	< 0.5					
Max. AC fall protection (A)	35					
Switch on current (A) / duration (ms)	< 5 / < 40					
PROTECTION, ENVIRONMENTAL CONDITIONS						
Cooling	Natural convection					
Ambient temperature (°C)	-25 ... +55					
Storage temperature (°C)	-25 ... +55 (according to IEC 60721-3-1 1K4)					

Type	08K	10K	13K	17K	20K	23K-MV
Item no.	867P008	867P010	867P013	867P017	867P020	867P023
Transport temperature [°C]	-25 ... +70 (according to IEC 60721-3-2 2K3)					
Relative ambient humidity [%]	0 ... 100					
Site altitude [m above NN]	3,843*					
Noise level [dBA]	< 45					
Internal overvoltage protection (EN 61643-11)	Type 3					
Protection class (IEC 62103)	I					
Overvoltage protection (EN 60664-1)	DC: II, AC: III					
Environmental classes (IEC 721-3-4)	4K4H					
Degree of pollution (IEC 72162109-1)	III					
Certification	Current certificates can be found at www.refu.com					
SZS or grid protection	according to DIN VDE V 0126-1-1					
GENERAL DATA						
Interfaces	Ethernet, RS485, irradiation and temperature sensor					
Protection class (IEC 60529)	IP65					
Dimensions W x H x D (mm)	535 x 601 x 277					
Dimensions with packaging W x H x D (mm)	595 x 636 x 292					
Device weight (kg)	38.4					
Weight with packaging (kg)	43.5					

* Note derating of max. DC voltage:

Amount over NN.	Max. DC voltage
Up to 3,843 m	1,000 V
Up to 4,000 m	971 V

8 Contact

Please address any questions on malfunctions or technical problems to:

Service-Hotline: +49 (0)7121 4332 - 333
(Monday - Thursday, 8 am to 5 pm, Friday 8 am to 4 pm)

Online

Email: service@refu.com

Website: www.refu.com

You should have the following data at hand:

- Exact description of the error with error code
- Device type
- Serial number
- Firmware version

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